**Name:** Key ______ / 25

**Please circle your instructor's name:** Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. **[10 points]** You are asked to contruct a right circular cylinder with radius $r$ and height $h$ such that the sum of the radius and the height is constant and equal to 6 inches, i.e.

$$r+h=6. \quad \Rightarrow \quad h = 6-r$$

Determine the dimensions of such a cylinder that produce the **maximum volume**. Note that the volume $V$ of a cylinder is given by

$$V=\pi r^2 h.$$

Find the values of $r$ and $h$ that maximize $V$, and justify that those dimensions give the maximum volume.

$V = \pi r^2(6-r) = 6\pi r^2 - \pi r^3$ Domain $r \in [0,6]$

$V' = 12\pi r - 3\pi r^2 = 3\pi r(4-r)$

Critical points: $r = 0, 4$

| $r$ | $V$ |
|---|---|
| 0 | 0 |
| 4 | $32\pi$ |
| 6 | 0 |

absolute max

Dimensions that maximize $V$ are $\underline{r = 4 \text{ inches}}$, $\underline{h = 6-4 = 2 \text{ inches}}$

**2. [8 points]** Evaluate the following limits. **Show your work**, uncluding appropriate use of limit notation. If you use L'Hôpital's rule, you must indicate where you are using it by writing $\overset{H}{=}$ or $\overset{L'H}{=}$ or something similar. Use $\infty$ or $-\infty$ where appropriate, and if the limit does not exist, write DNE and provide a justification.

**a.** $\lim\limits_{t\to 2} \dfrac{e^{t-2}-t+1}{t^2-4t+4} \overset{L'H}{=} \lim\limits_{t\to 2} \dfrac{e^{t-2}-1}{2t-4} \overset{L'H}{=} \lim\limits_{t\to 2} \dfrac{e^{t-2}}{2} = \boxed{\dfrac{1}{2}}$

I.F. $\frac{0}{0}$ I.F. $\frac{0}{0}$

**b.** $\lim\limits_{x\to 0^+} 2x\ln(3x) = \lim\limits_{x\to 0^+} \dfrac{2\ln(3x)}{x^{-1}} \overset{L'H}{=} \lim\limits_{x\to 0^+} \dfrac{\frac{2}{3x}\cdot 3}{-x^{-2}} = \lim\limits_{x\to 0^+} -x^2\left(\dfrac{2}{x}\right) = \lim\limits_{x\to 0^+} -2x = \boxed{0}$

I.F. $\frac{\infty}{\infty}$

**3. [7 points]** Evaluate the following indefinite integrals (antiderivatives):

**a.** $\displaystyle\int \left(\sqrt{x}+4e^x-\sin(x)\right)\,dx = \boxed{\frac{2}{3}x^{\frac{3}{2}} + 4e^x + \cos x + C}$

**b.** $\displaystyle\int \frac{2x^5-x^2+3}{x^2}\,dx = \int\left(2x^3 - 1 + \frac{3}{x^2}\right)dx = \boxed{\frac{1}{2}x^4 - x - 3x^{-1} + C}$